

Advanced AI System (C++)

A highly modular **AI Shooter Framework built entirely in C++ for Unreal Engine 5**, designed as a foundation for tactical, stealth, and action-oriented shooter games. The system combines intelligent AI behavior with complete combat, weapon, damage, locomotion, perception, and gameplay systems.

Key Features

- **Advanced AI Combat**

- Reactive AI shooting with support for different weapon types.
- Intelligent cover selection with peek and shoot behavior.
- Stealth attacks and silent takedown mechanics.
- AI and player reload systems.

- Healing system with Niagara visual effects and audio feedback.

- **AI Behavior & Perception**

- Patrol system supporting random and predefined patrol routes.
- Investigation system for responding to sounds and disturbances.
- Last-known-location tracking when the player leaves line of sight.
- AI Perception integration for **Sight, Sound, and Damage**.
- Behavior Tree and Blackboard architecture for flexible AI customization.

- **Multiple AI Archetypes**

- **Aggressive Shooter AI** — Rushes, flanks, and maintains pressure on the player.
- **Stealth Shooter AI** — Prioritizes cover, positioning, and controlled engagements.
- **Pistol Shooter AI** — Optimized for medium-range combat.
- **Rifle Shooter AI** — Designed for long-range tactical engagements and line-of-sight management.
- Extensible architecture for creating additional AI variants.

- **Advanced Locomotion**

- Shared locomotion framework for both player and AI characters.
- Walking, running, crouching, jumping, and directional aiming.
- Smooth movement transitions and animation integration.
- Support for Animation Blueprints and Montages.

- **Modular Damage System**

- Supports ranged, melee, radial, and directional damage.
- AI reactions based on different damage sources.
- Directional damage indicators for player feedback.
- Modular architecture that can be extended with additional damage types.

- **Dual Weapon System**

- Primary and secondary weapon slots.
- Equip, unequip, and weapon switching functionality.
- Support for multiple weapon types.
- Integrated animation and socket switching logic.

- **Weapon Selection Wheel**

- Radial weapon-selection interface.
- Smooth selection and transition behavior.
- Designed to be easily extended with additional weapons and slots.

Technical Focus

The framework is built with a strong focus on **modularity, scalability, and reusability**. Core systems are separated into extensible components and base classes, making it possible to create new AI behaviors, weapons, combat mechanics, and gameplay variations without rebuilding the entire system.

This project demonstrates experience with **Unreal Engine 5 C++ gameplay programming**, including AI Perception, Behavior Trees, Blackboard systems, combat mechanics, weapon handling, damage architecture, locomotion, animation integration, and modular gameplay design.